Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected missing-data validation record.

Distinguish zero and unknown, trace filtering, repair truthiness and explain which sessions are excluded.

Keep one habit: Retain original rows, accepted order, review reasons, field-specific bounds and reported denominators.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Noise and Outliers

Choose Informative Questions

Accuracy as a Fraction

W1 · Distinguish zero, absent and None

> Explain what is measured and what is unknown.

Your evidence record

- Duration reference: exact Python int, minutes 0–60 inclusive; None or absent minutes is unknown, and Boolean is not a count. Original rows in order:0,absent,12,-1,None. W2 supplies a separate four-value list. The original filter accepts2/5; the extended lab adds60,text"5",True and accepts3/8. Do not mix the separate stock maximum 8 or independent maximum 10 into this duration rule.

- Explain the difference between {"minutes": 0}, {} and {"minutes": None}.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace four supplied durations

> Keep exact type, range, order and rejection count.

Your evidence record

- Duration reference: exact Python int, minutes 0–60 inclusive; None or absent minutes is unknown, and Boolean is not a count. Original rows in order:0,absent,12,-1,None. W2 supplies a separate four-value list. The original filter accepts2/5; the extended lab adds60,text"5",True and accepts3/8. Do not mix the separate stock maximum 8 or independent maximum 10 into this duration rule.

- Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair a truthiness condition

> Give a valid zero and invalid nonempty-text counterexample.

Your evidence record

- Duration reference: exact Python int, minutes 0–60 inclusive; None or absent minutes is unknown, and Boolean is not a count. Original rows in order:0,absent,12,-1,None. W2 supplies a separate four-value list. The original filter accepts2/5; the extended lab adds60,text"5",True and accepts3/8. Do not mix the separate stock maximum 8 or independent maximum 10 into this duration rule.

- A program uses if value to accept durations. Give a counterexample and the correct rule.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Report rejected-record coverage

> Use the original five-record population in this question.

Your evidence record

- Duration reference: exact Python int, minutes 0–60 inclusive; None or absent minutes is unknown, and Boolean is not a count. Original rows in order:0,absent,12,-1,None. W2 supplies a separate four-value list. The original filter accepts2/5; the extended lab adds60,text"5",True and accepts3/8. Do not mix the separate stock maximum 8 or independent maximum 10 into this duration rule.

- Only 2 of 5 supplied durations were accepted. Why should the report include this?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Validate independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Independent policy: exact int only, inclusive count bounds 0–10; preserve input order and hold missing/type/range failures for review. Quote dictionary keys in Python.

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Validate independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace filtering for **Missing value**s and **validation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Missing value**s and **validation**. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 18 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

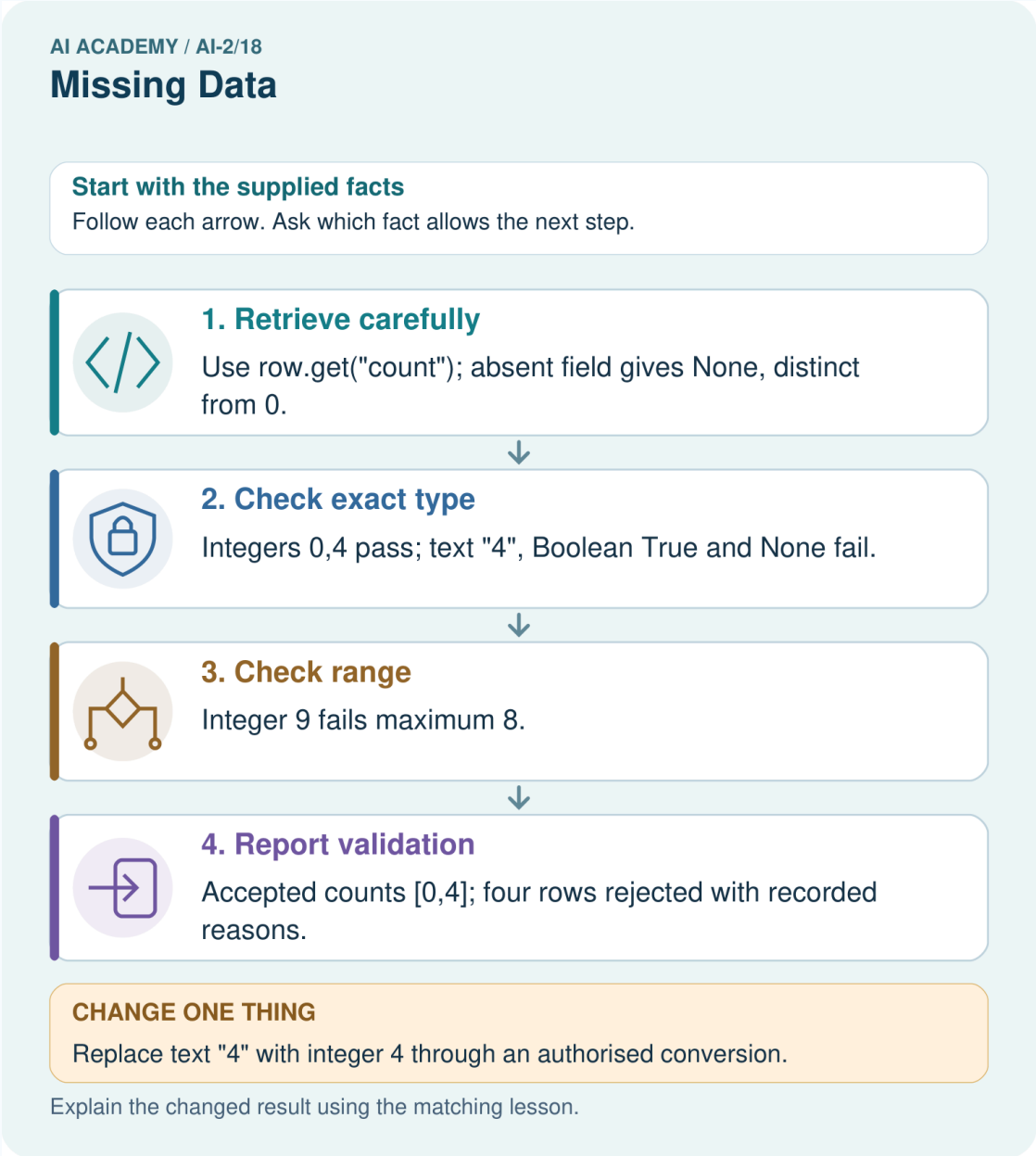
> Keep the explanation and the support you used.

Use this when helpful

A fictional catalogue records counts of available copies under a strict **input** contract.

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional catalogue records counts of available copies under a strict **input** contract.
- Which labelled step matches this detail? Accepted counts [0,4]; four rows rejected with recorded reasons.
- 1. Report **validation**
- 2. Retrieve carefully

- 3. Check range
- 4. Check exact type
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report **validation** Accepted counts [0,4]; four rows rejected with recorded reasons. Testing if value would wrongly discard zero, which is a real valid count.

Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Changed-case question

- Changed-case practice: Replace text "4" with integer 4 through an authorised conversion. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Accepted values become [0,4,4]; three rows remain rejected.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Repair and save your work

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Missing Data: the case

AI ACADEMY / AI-2/18 · 01 · LOOK AT THE FACTS

Missing Data: the case

The situation

A fictional catalogue records counts of available copies under a strict input contract.

Supplied evidence

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.
Rows have count 0, count 4, absent count, count "4", count True and count 9.

By the end, I can validate missing values without confusing zero with absent data.

Follow the labelled picture with the matching explanation. Open the vector diagram

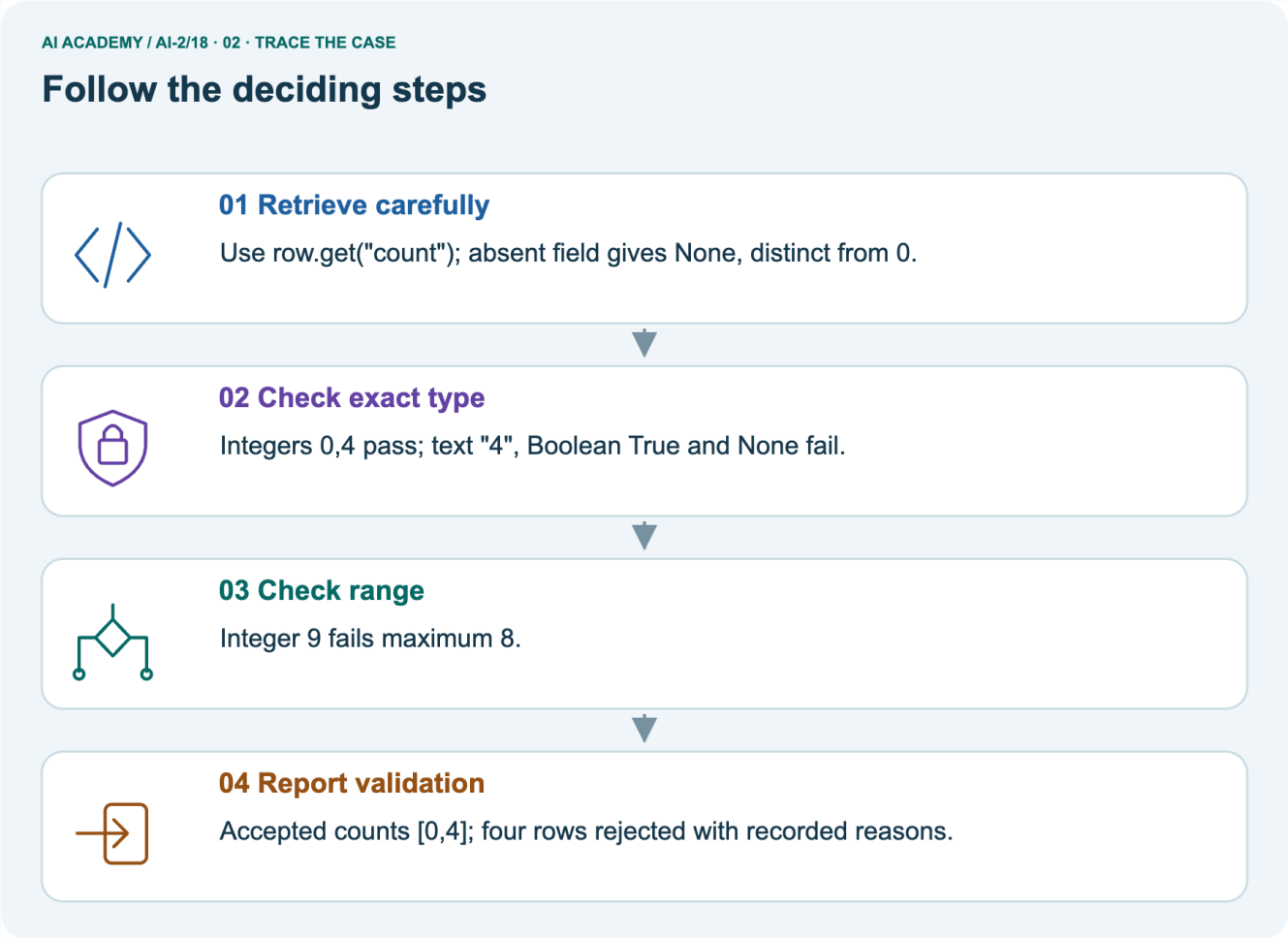
A fictional catalogue records counts of available copies under a strict **input** contract.
Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

AI ACADEMY / AI-2/18 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Recorded zero	0	Accept as a measurement
Absent or None	Unknown	Hold for review
Negative integer	Outside range	Reject under this schema

Read each row from left to right and connect the evidence to the stated rule.

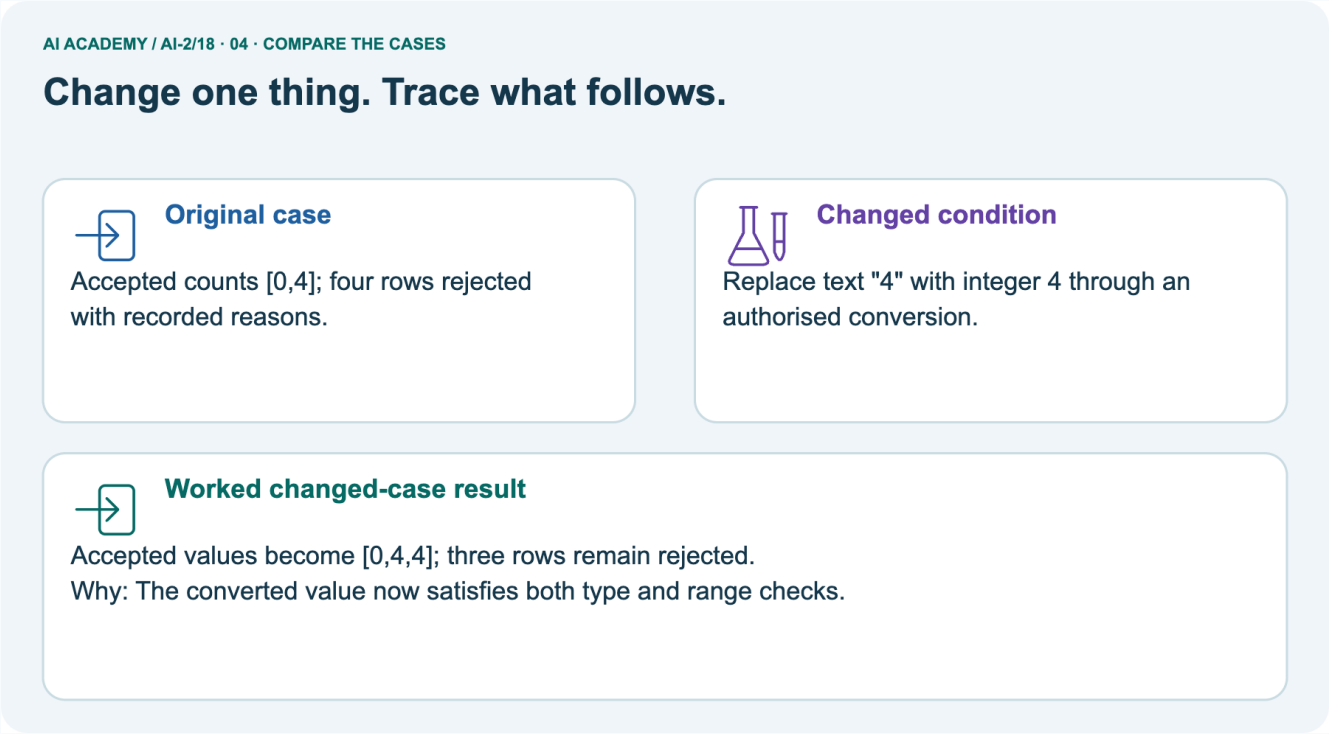
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

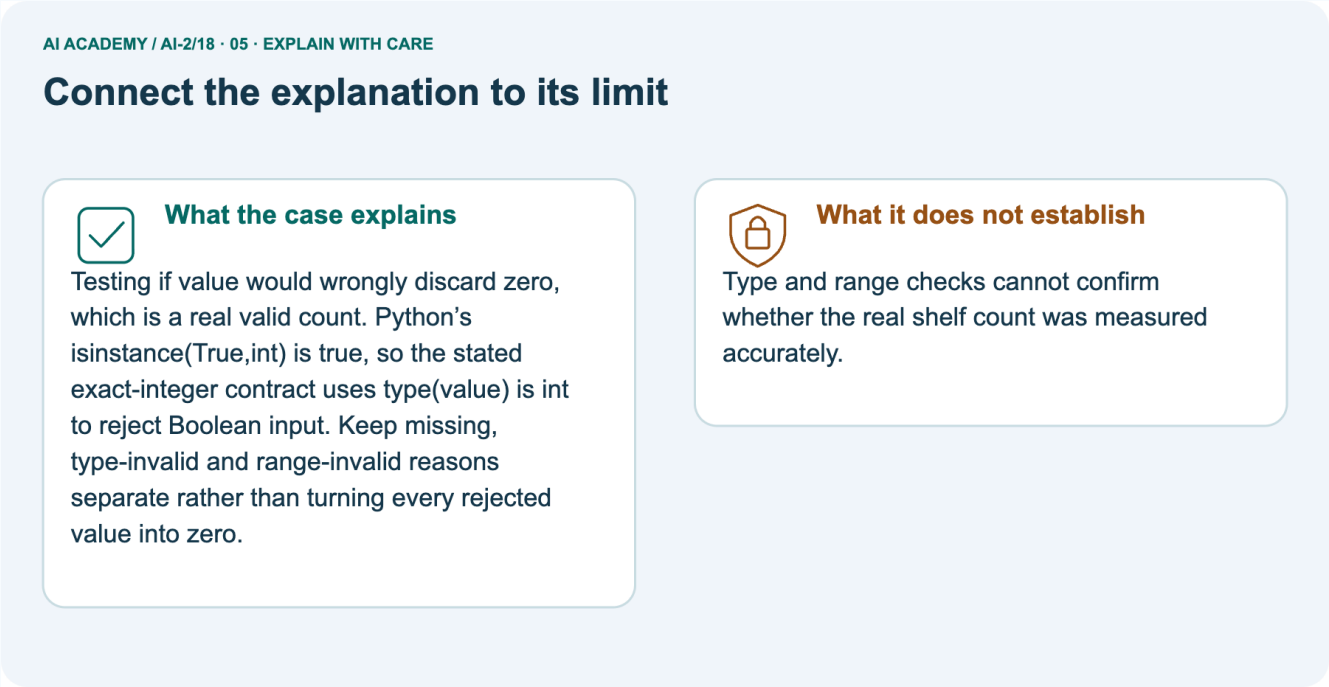
Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

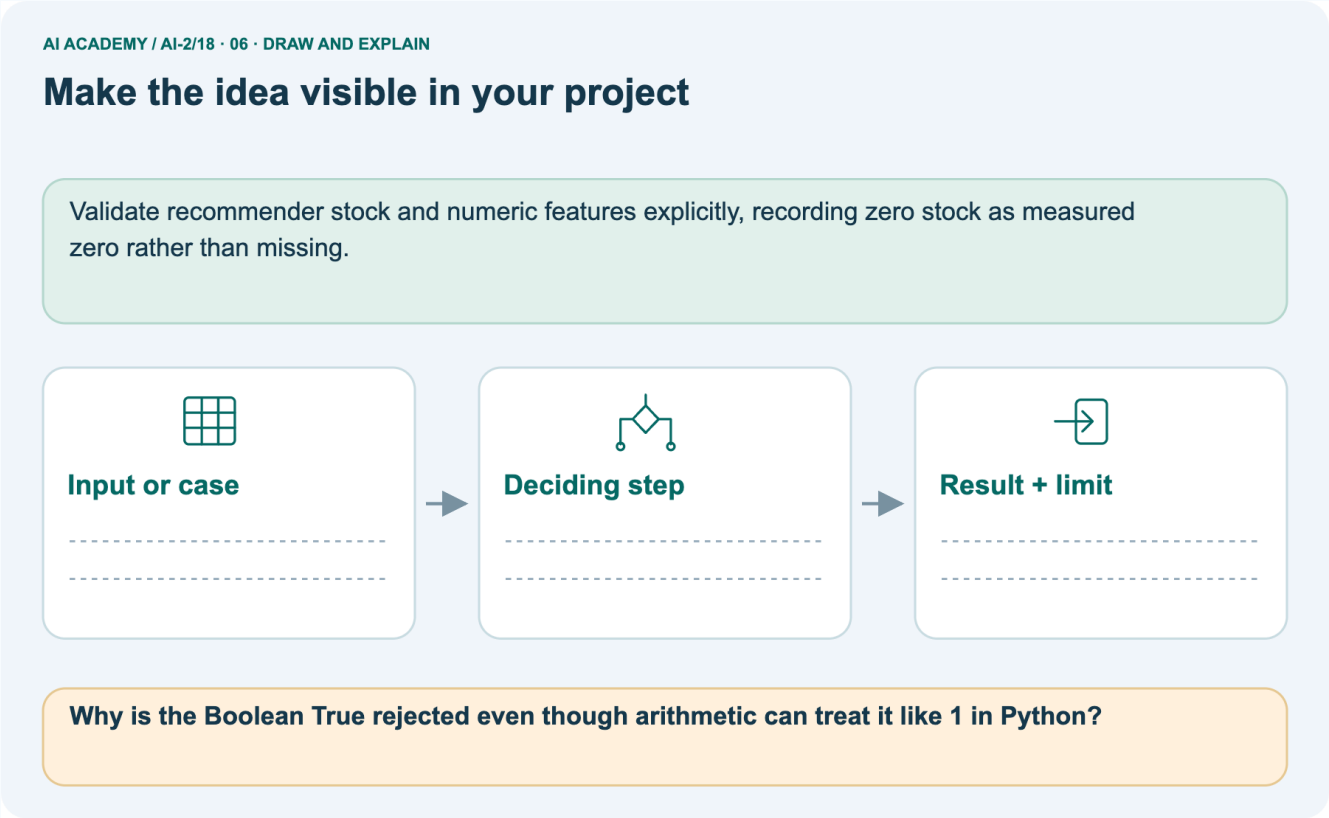
Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Missing Data: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.